Inverted MAML Card 1: Round 7 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=2048, Mom=0.85, Clamp=±75 LSB

1. Round 7 Summary & Cumulative Comparison (R1–R7)

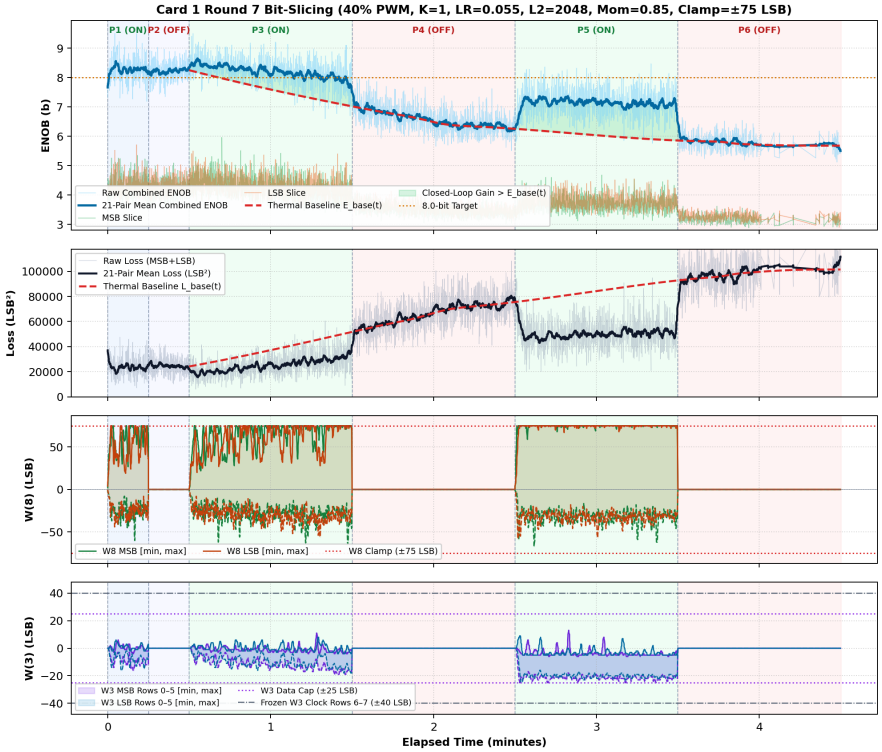
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 7 Detrended Trend Metrics:** Lift vs E_base = **+1.04 b**, Loss Ratio = **1.74x**, Climb = **+1.17 b**, Jitter = **0.40 b**, Recovery = **0.50**, Score = **+2.665**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1dc_fix	0.03	1024	0.9	±75	+1.00b	1.71x	+0.95b	0.40b	0.00	+1.799
R2dc_fix	0.035	1536	0.85	±75	+0.99b	1.69x	+1.16b	0.39b	0.00	+1.957
R3dc_fix	0.02	2560	0.8	±70	+0.04b	1.00x	+0.24b	0.34b	1.00	+0.204
R4dc_fix	0.05	1792	0.85	±70	+1.08b	1.78x	+0.97b	0.40b	0.50	+2.480
R5dc_fix	0.055	1792	0.85	±85	+1.12b	1.77x	+1.37b	0.42b	0.00	+2.393
R6dc_fix	0.055	1792	0.8	±90	+0.99b	1.72x	+1.02b	0.40b	0.00	+1.770
R7dc_fix*	0.055	2048	0.85	±75	+1.04b	1.74x	+1.17b	0.40b	0.50	+2.665

2. Phase-by-Phase Telemetry (3,806 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	313	8.04->8.28 (8.26b)	-	28591->22925	[75,75]	16	313
P2	Cold / Regler OFF	311	8.10->8.26 (8.22b)	-	27720->23924	[0,0]	0	0
P3	Warm / Regler ON	1052	8.43->8.02 (8.21b)	+0.79 b	19713->29585	[75,75]	19	1052
P4	Warm / Regler OFF	891	7.15->6.35 (6.58b)	+0.03 b	50545->73468	[0,0]	0	0
P5	Warm / Regler ON	794	6.43->7.08 (7.13b)	+1.18 b	71936->51348	[75,75]	25	794
P6	Warm / Regler OFF	445	5.93->5.70 (5.79b)	+0.00 b	90958->102041	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 7

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 7 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

